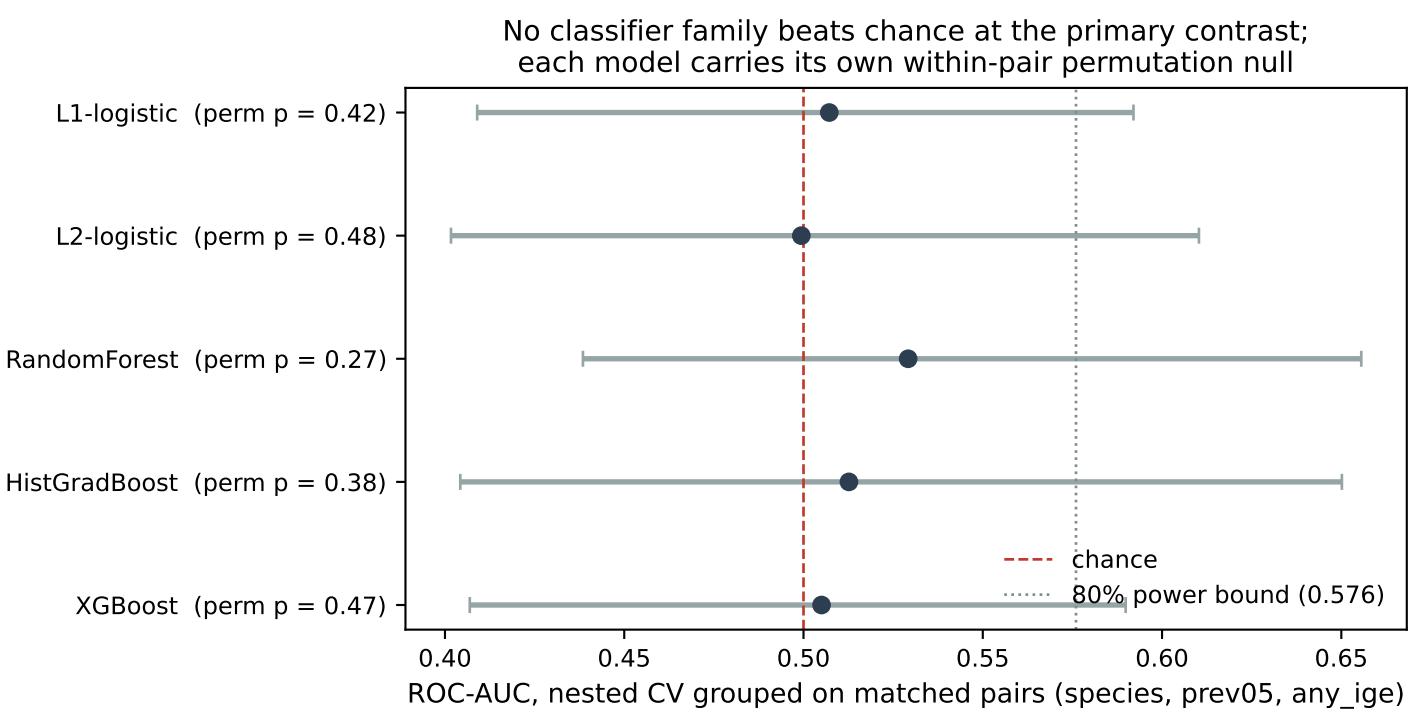
No classifier family beats chance at the primary contrast;
each model carries its own within-pair permutation null
L1-logistic (perm p = 0.42)
L2-logistic (perm p = 0.48)
RandomForest (perm p = 0.27)
HistGradBoost (perm p = 0.38)
XGBoost (perm p = 0.47)
chance
80% power bound (0.576)
0.40
0.45
0.50
0.55
0.60
0.65
ROC-AUC, nested CV grouped on matched pairs (species, prev05, any_ige)